

Climate Change Impact on Agriculture – Data Analysis and Prediction

1. Abstract

Climate change is one of the most significant global challenges affecting agriculture. Variations in temperature, rainfall, and other environmental factors directly influence crop productivity. This project analyses historical climate and agricultural data to understand the relationship between climate variables and crop yield. Using data analysis techniques and machine learning models, the project predicts crop yield based on climatic conditions. The results are stored in a MySQL database and can be visualized using dashboards such as Power BI for better decision-making.

2. Introduction

Agriculture plays a crucial role in the global economy and food security. However, climate change has introduced unpredictable weather patterns, including rising temperatures, irregular rainfall, and extreme weather events. These changes affect soil quality, water availability, and crop productivity.

Data analysis and machine learning techniques help in identifying patterns between climatic factors and agricultural output. Predictive models can assist farmers, policymakers, and researchers in understanding how climate conditions influence crop yield and in planning agricultural strategies accordingly.

3. Goal of the Project

The main objectives of this project are:

- To analyse climate and agricultural datasets.
- To study the relationship between temperature, rainfall, and crop yield.
- To build machine learning models for crop yield prediction.
- To compare different prediction models.
- To store prediction results in a MySQL database.
- To prepare data for visualization in dashboards such as Power BI.

4. Dataset Overview

The dataset used in this project contains 350 records related to climate conditions and agricultural yield. The dataset includes several attributes such as:

- Year
- Average Temperature (°C)
- Total Precipitation (mm)
- Humidity (%)
- Soil Moisture
- Crop Yield (MT per hectare)

These variables help analyse how environmental conditions affect crop production.

Dataset size: 350 rows and 11 columns

5. Tools and Technologies Used

Programming Language Python

Libraries Used • Pandas – Data manipulation and analysis • NumPy – Numerical computation • Matplotlib – Data visualization • Seaborn – Statistical visualization • Scikit-learn – Machine learning models

Database MySQL – Used to store prediction results

Visualization Tool Power BI – Used to build interactive dashboards

Development Environment Visual Studio Code

6. Data Preprocessing

Before analysis, the dataset was cleaned and prepared.

Removing Duplicates

Duplicate rows were removed to avoid biased results.

Handling Missing Values

Missing numerical values were replaced with the mean of the column, while categorical values were replaced with the most frequent value.

Encoding Categorical Data

Categorical variables were converted into numerical form using Label Encoding to make them suitable for machine learning models.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand patterns and relationships in the data.

Temperature Distribution

A histogram was used to visualize how temperature values are distributed across the dataset.

Rainfall vs Crop Yield

A scatter plot was created to analyse how precipitation levels influence crop yield.

Correlation Analysis

A heatmap was generated to identify relationships between different variables in the dataset.

This analysis helps identify important climate factors affecting crop productivity.

8. Machine Learning Models

Two machine learning models were used to predict crop yield.

1. Linear Regression

Linear Regression is a simple regression model that predicts the relationship between dependent and independent variables using a linear equation.

Result: R^2 Score $\approx$ 0.56

2. Random Forest Regressor

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy.

Result: R^2 Score $\approx$ 0.54 RMSE $\approx$ 0.73

Random Forest provides stable predictions and works well with complex datasets.

9. Model Evaluation

The models were evaluated using the following metrics:

R^2 Score Measures how well the model explains variance in the data.

Root Mean Square Error (RMSE) Measures the average prediction error.

The results show that both models perform reasonably well for predicting crop yield based on climate conditions.

10. Database Integration (MySQL)

After generating predictions, the results were stored in a MySQL database.

Database Name agriculture_db

Table Name crop_yield_predictions

Table Structure:

- id – Unique record identifier
- actual_yield – Actual crop yield value
- predicted_yield – Model predicted yield

Approximately 70 prediction records were stored in the database, representing the test dataset used for model evaluation.

11. Visualization

The stored prediction data can be connected to Power BI to create dashboards such as:

- Actual vs Predicted Crop Yield
- Crop Yield Distribution
- Climate Factor Impact Analysis
- Prediction Error Analysis

These dashboards help stakeholders understand agricultural trends and climate impacts.

12. Results and Findings

- Climate variables such as temperature and rainfall significantly influence crop yield.
- Machine learning models can predict agricultural productivity with moderate accuracy.
- Random Forest provided stable predictions for this dataset.
- Integrating Python with MySQL allows efficient storage and retrieval of prediction results.
- Visualization tools like Power BI help present insights clearly.

13. Conclusion

This project demonstrates how data analysis and machine learning techniques can be used to understand the impact of climate change on agriculture. By analyzing climatic factors and predicting crop yield, the system provides useful insights for agricultural planning and decision-making. The integration of Python, MySQL, and visualization tools makes it possible to build a complete data analytics pipeline for agricultural analysis.

14. Future Improvements

- Use larger and real-time climate datasets.
- Apply advanced models such as Gradient Boosting or XGBoost.
- Integrate live weather APIs for real-time predictions.
- Deploy the system as a web application for farmers and policymakers.
- Build advanced dashboards for monitoring agricultural trends.

15. References

- Scikit-learn Machine Learning Documentation
- Pandas Data Analysis Library
- MySQL Database Documentation
- Research articles on climate change and agriculture